

LRMI calculation pipeline — methods summary

Lithuanian River Macroinvertebrate Index from the Operational Taxalist (OTL)

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Yellow-highlighted values in this document mark thresholds, anchors, counts and other numerical choices that should be verified against the source paper, the LEPA reference, or the raw data before publication.

1 Purpose and scope

This pipeline takes raw macroinvertebrate sampling data from Lithuanian rivers, harmonises taxonomy through a project-specific Operational Taxalist (OTL), and recalculates the multimetric **Lithuanian River Macroinvertebrate Index (LRMI)** under the EU Water Framework Directive (WFD). The recalculated values are then compared against the LEPA-published EQRs and EQCs for a curated subset of sites to test whether OTL standardisation materially alters the assessment.

The pipeline is implemented as seven core R scripts in **Scripts/** (numbered 1–7) plus three diagnostic scripts (8–10) and a self-contained HTML decision tree. Inputs sit in **Inputs/**; every output is written to **Outputs/** prefixed with the **producing script’s number** so the mapping is one-to-one (e.g. script 6 writes **Outputs/6_LRMI_with_comparison.xlsx**).

Scope: all analyses from Stage 4 onwards are restricted to **2014–2022**. Pre-2014 LEPA EQRs were calculated using the older single-DSFI assessment rather than the LRMI, and including them in the comparison would conflate methodological era with taxonomic standardisation effects (see Findings, §10).

2 Workflow at a glance

	Stage	Script	Purpose
	1	1_generate_decision_tree_html_v17.R	Render decision-tree HTML from editable text blocks
	2	2_OTU_assignment_v22.R	Apply decision tree to assign an OTU to every taxon
	3	3_DSFI_calculator_v15.R	Compute the Danish Stream Fauna Index (DSFI) per sample

Stage	Script	Purpose
4	4_calculation_of_taxonomic_indices.R	Compute BMWP / ASPT, group richness/abundance, compound indices
5	5_merge_indices_and_metadata.R	Join DSFI + taxonomic indices + site metadata
6	6_calculate_LRMI.R	Calculate the LRMI and ecological-quality class per site-year
7	7_comparing_eqrs.R	Compare LRMI / class against LEPA's published values
8	8_DEP_OTL_vs_validated.R (diagnostic)	Test whether residual LEPA-vs-LT differences trace to #DEP
9	9_per_metric_diagnostic.R (diagnostic)	Per-metric breakdown for class-mismatch site-years
10	10_lepa_per_metric_comparison.R (diagnostic)	Direct head-to-head against LEPA's published per-metric EQRs

3 Stage 1 — Decision-tree design (Script 1)

3.1 Inputs to the design

A Lithuanian aquatic-invertebrate species list was compiled from national literature and validated against the GBIF and Molluscabase taxonomic backbones. Four specialists (2 current EPA biomonitoring identifiers, 1 former identifier now calculating national EQRs, 1 independent researcher) were asked to state, for each species, the **coarsest** taxonomic level they could reliably identify it to (species / genus / subfamily / family / order / subclass / class / phylum).

3.2 Three synchronised representations

To keep the rules transparent and reviewable, the decision tree is maintained in three locked-step representations:

- `Outputs/1_OTL_decision_tree_v17_machine_readable.md` — the canonical human-readable specification (plain text, version-controlled).
- `Outputs/1_OTL_decision_tree_v17.html` — visual flow-chart for non-technical reviewers and supplementary material.
- The R script `1_generate_decision_tree_html_v17.R` — generates the HTML directly from editable content blocks, so the visual document cannot drift out of sync with the rule text it represents.

Any rule change is made in the `.md` (authoritative) and the R generator script in parallel; sourcing the generator re-renders the HTML. The four-phase logic below is the content these three representations agree on.

3.3 OTU assignment rules

The OTU for each taxon is determined by a four-phase decision tree (see [Outputs/1_OTL_decision_tree_v17.html](#) for the full visual version):

Phase 1 — Preparation. Validate names, flag exclusions from EQR using the `excluded_taxa` sheet as the source of truth.

Phase 2 — Forced-level overrides. A small number of groups receive a fixed OTU level regardless of specialist agreement. These taxa **skip Phases 3 and 4** entirely. The overrides are:

Group	Forced level	EQR?	Reason
Entirely excluded groups (all families excluded)	group level	excluded	non-target / non-aquatic / non-macroscopic
Oligochaeta (Naididae)	family	included	DSFI uses Naididae at family (IG6)
Other Oligochaeta	subclass	included	difficult routine identification
Neuroptera (Osmylidae, Sisyridae)	family	included	specialist order-level was too coarse
Stenostomidae	phylum	excluded	order is taxonomically unstable
All other excluded-from-EQR taxa	family	excluded	natural ceiling at family

Entirely excluded groups:

- Phyla: Porifera, Cnidaria, Nematoda, Nematomorpha, Nemertea, Bryozoa
- Classes: Polychaeta, Arachnida, Collembola, Maxillopoda, Branchiopoda
- Orders: Stylommatophora, Dolichomicrostomida, Prolecithophora, Rhabdocoela, Anostraca

Phase 3 — 100% specialist agreement. For each remaining species-level taxon, the OTU is set to the **coarsest** level reported by any of the four specialists (i.e. the level all four can reach).

Phase 4 — Metric compliance. If the Phase-3 level is coarser than the minimum required by any downstream metric (DSFI, BMWP, etc.), the OTU is overridden to the minimum required level. Overrides are documented row-by-row in the output trace.

3.4 Truly-aquatic rule for higher-level entries

Order-and-above entries are retained for EQR calculation only for **truly aquatic** groups: Bivalvia (class); Plecoptera, Ephemeroptera, Odonata, Megaloptera, Trichoptera (orders). All other order+ entries are excluded from EQR (they may include terrestrial representatives).

4 Stage 2 — OTU assignment (Script 2)

2_OTU_assignment_v22.R implements the decision tree above against the project's three reference sheets:

- `Specialist_taxalist` — one row per validated species, with each specialist's reported identification level.
- `OTL_final` — the full universe of taxa (species, genera, families and higher-level entries).
- `excluded_taxa` — the source of truth for which taxa are excluded from EQR calculations (entirely-excluded groups + individually excluded families/species).

The script outputs an Excel workbook with three sheets:

1. `OTL_final_with_OTU` — every row of `OTL_final` annotated with its assigned OTU name, OTU level, metric group, and EQR-exclusion flag.
2. `decision_tree_trace` — full per-taxon explanation of which phase triggered, what the agreement level was, and whether a metric override was applied. This is the audit trail for the supplementary material.
3. `hierarchical_OTL` — a parallel taxalist with fallback entries at every level from phylum down to species, so damaged or juvenile specimens identifiable only to family or order still have a valid OTU.

4.1 Data-quality guard

After reading the input sheets, the script warns when a name appears in one sheet but not another. This was added after discovering that a typo (*Aelosoma hemprichi* in `Specialist_taxalist` vs *Aeolosoma hemprichii* in `excluded_taxa`) was silently breaking the class-level Polychaeta exclusion.

Output: `Outputs/2_OTU_assignments_v22.xlsx`.

5 Stage 3 — DSFI calculation (Script 3)

Implements Skriver et al. (2000) for each sample (`site_code` × `year`). The script aggregates abundances to DSFI-relevant taxa, applies presence-threshold rules, then walks through six Indicator Groups (IG1–IG6) to look up the DSFI value (1–7) in the published Table 2 matrix.

Key presence rules:

- Kick sample threshold: 2 individuals for indicator-group determination (Table 2). Hand-picked sample threshold: 1. Diversity groups (Table 3) ignore the kick-2 filter — 1 specimen anywhere is enough.
- Oligochaeta as a negative diversity group requires 100 individuals in the kick sample.
- Indicator-group precluders: Asellus kick 5 → IG3; Chironomus kick 5 → IG4; Eristalini kick 2 → IG6.
- IG5 lower row used when Oligochaeta kick 100.
- Simuliidae presence at IG5 requires kick 25 or hand 1.

Output: `Outputs/3_DSFI_results_all_sites.xlsx` and a curated subset of 95 sites at `Outputs/3_DSFI_results_subset.xlsx`.

6 Stage 4 — Taxonomic indices (Script 4)

Computes the remaining LRMI inputs and a wider set of community metrics from the OTL-standardised data:

- **Total richness, abundance**, Margalef / Shannon / Simpson / Pielou (not all used downstream; retained for completeness).
- **BMWP and ASPT** — sum of family-level sensitivity scores ÷ number of scoring units. Two scoring exceptions:
 - *Ancylus* sp. is scored as its own **genus-level** unit (the family Ancyliidae was synonymised into Planorbidae but retains a distinct BMWP score).
 - **Oligochaeta** collapses to a single unit at score 1, regardless of how many oligochaete OTUs are present (Naididae, Tubificidae, Oligochaeta Gen. sp.).
- **Group richness / abundance** for individual orders (Ephemeroptera, Plecoptera, Diptera, Hemiptera, Crustacea, ...) and for compound groups (CrHi, EHP, DEP).
- **DEP richness** = number of unique Diptera families + number of individual E and P taxa at species level (the LEPA convention).
- **%EHP – %CrHi** = $(\sum \text{EHP abundances} - \sum \text{CrHi abundances}) / \text{total abundance}$.

6.1 Hand-picked abundance treatment

Kick-sample abundances are kept as raw counts. **Hand-picked abundances are coerced to 1 before aggregation** (presence-only) and then summed with the kick counts within each `site_id × year × OTL_taxonname` group. The rationale is that the hand-picked sub-sample is qualitative — it is meant to capture taxa that the kick sample may have missed, not to provide quantitative density estimates. Treating hand-picked as presence/absence prevents the qualitative supplement from skewing abundance-weighted metrics (BMWP, %EHP – %CrHi, group abundances).

6.2 Year filter

From this stage onwards the analysis is restricted to 2014–2022. The filter is applied at the same point as the existing taxon-exclusion filters (NO MATCH, EXCLUDE, site_code prefix LTR). Rationale is in Findings (§10).

Output: `Outputs/4_calculated_taxonomic_indices.xlsx`.

7 Stage 5 — Merge with site metadata (Script 5)

5_merge_indices_and_metadata.R joins:

1. DSFI results on (site_id, site_code, year).
2. Taxonomic indices on the same key.
3. **Site metadata** (site_meta_data sheet) on site_id — static per-site fields: location, waterbody type, river basin, river type, modification state, reference-site flag.
4. **EQR metadata** (first 20 columns of clean_macroinvertebrate_data) on (site_id, site_code, year) — sample-level fields including LEPA’s published EQR and EQC.

The EQR metadata sheet uses 0 and "0" as placeholders for “never assessed”. These are converted to NA at load time, then rows with NA in either EQR or EQC are dropped, so all downstream comparisons rest on genuine LEPA assessments only.

The same 2014–2022 year filter is applied to DSFI, taxonomic- indices, and EQR-metadata inputs on read. The site_meta_data sheet (static per-site fields, no year column) is not filtered.

Output: Outputs/5_merged_indices.xlsx.

8 Stage 6 — LRMI calculation (Script 6)

Implements Sidagytė-Copilas & Arbačiauskas (2022).

8.1 The four core metrics and their anchors

For each metric i : $EQR_i = (x_i - lower_i) / (ref_i - lower_i)$.

Metric	Reference value	Lower anchor
DSFI	7	1
ASPT	7.0	1.0
#DEP	15	0
%EHP – %CrHi	0.6	-1.0

8.2 LRMI formula and our strict variant

The published method averages all four EQRs, falling back to the remaining three if DSFI fails to key out. **We have chosen to be stricter:** if any of the four metrics is NA, LRMI is NA. This avoids reporting an LRMI built on only a subset of metrics.

8.3 Clamping and rounding (project convention)

Per-metric EQRs are kept unclamped and unrounded as internal inputs. The final LRMI is then:

1. Computed as the arithmetic mean of the four raw EQRs (`LRMI_raw`).
2. Clamped at **1** (no LRMI > 1).
3. Rounded to **2** decimal places (matches LEPA’s reported precision).

Both `LRMI_raw` (faithful to the paper) and `LRMI` (display / comparison value) are retained in the output.

8.4 Ecological class boundaries

Uniform across all river types, derived from the clamped + rounded LRMI (matches LEPA’s class-from-rounded-EQR convention):

Class	Range
High	LRMI 0.80
Good	0.60 LRMI < 0.80
Moderate	0.40 LRMI < 0.60
Poor	0.30 LRMI < 0.40
Bad	LRMI < 0.30

Type-specific adjustments proposed by Sidagytė-Copilas & Arbačiauskas (2022, their Table 5, italic rows) are **not** applied — the uniform “Current” boundaries match LEPA’s operational practice.

8.5 Natural-waterbody scope

Only sites with `modification_state == "N"` (natural) enter the LRMI calculation. Heavily modified or artificial waterbodies use a different reference-condition formulation that is out of scope for this project.

9 Stage 7 — Comparison against LEPA (Script 7)

9.1 Curated comparison subset

A **27**-site subset covering a POOR → HIGH gradient was selected, with **2–3** samples per site over **2015–2022** (**82** site-years total). Sites are distributed across the four main Lithuanian catchments and across all five WFD river types.

9.2 What is compared

For each site-year in the subset, the LEPA-published EQR and EQC (from the comparison input) are paired with our recalculated LRMI and LRMI_class (from Stage 5). Comparisons include:

1. **1:1 scatterplot** of LEPA EQR vs LT-OTL LRMI, coloured by LEPA EQC using the WFD palette, with Pearson correlation.
2. **Bland–Altman plot** with 95% limits of agreement ($\text{mean} \pm 1.96 \times \text{SD of differences}$) to surface systematic bias or wide spread.
3. **Boxplots by river type**, with horizontal WFD class bands as visual context.
4. **Sample-level table** with every site-year’s signed `eqr_diff` and a boolean `eqc_match` flag, sorted by absolute EQR difference so the largest discrepancies surface at the top. No banded categories are applied; the continuous residual is reported as-is.

9.3 Outputs

- `Outputs/6_LRMI_with_comparison.xlsx` (from Stage 6) — all site-years, full metric breakdown, both raw and clamped+rounded LRMI. Consumed by this stage.
- `Outputs/7_sample_level_comparison.xlsx` — per-sample table (LEPA vs LT-OTL) for case-by-case investigation, sorted by absolute EQR difference.
- `Plots/Figure_1.tiff` — map of Lithuanian river sampling sites coloured by latest-year EQC.
- `Plots/Figure_2.tiff` (script 6) — 1:1 scatter of LEPA EQR vs LT-OTL EQR with WFD class bands.
- `Plots/Figure_3.tiff` (script 10) — per-metric A/B comparison panel (scatter on top, box-plots on bottom).

The comparison input is also year-filtered to 2014–2022 at read time, defensively (the curated subset is already 2015–2022).

10 Stage 8 — Diagnostics

Two ad-hoc scripts that don’t feed downstream but ground the discussion of where LEPA-vs-LT residuals come from.

10.1 DEP-attribution diagnostic (script 8)

Script: `8_DEP_OTL_vs_validated.R`. Tests the hypothesis that residual LEPA-vs-LT differences come from OTL standardisation collapsing multiple species into a single OTU, artificially lowering the #DEP count. For every site-year, DEP is recomputed twice — once grouped by `OTL_taxonname` (matches our LRMI), once by `taxonname_validated` (pre-OTL). The implied LRMI shift from any DEP difference is then compared against the actual LEPA-vs-LT residual.

Headline result: the correlation between the DEP-implied LRMI shift and the actual residual is 0.17 — weak. Mean |residual| drops only from 0.0256 to 0.0241 after subtracting the DEP-implied shift. **OTL standardisation is not the dominant source of LEPA-vs-LT residuals.**

Output: `Outputs/8_DEP_OTL_vs_validated.xlsx`.

10.2 Per-metric diagnostic (script 9)

Script: `9_per_metric_diagnostic.R`. For a user-supplied set of site-years (the 46 sites with the largest residuals or class-mismatches), the four per-metric EQRs are pulled alongside their percentile rank against the full natural-waterbody distribution. The metric with the lowest percentile is flagged as the most-likely-culprit for each site.

Headline result: no single metric dominates. EHP-CrHi is the slight leader (17 / 46 sites flag it as the lowest metric; mean percentile 32.7 vs 40–42 for the others), but DSFI, ASPT, and DEP each drag down 9–10 sites. 40 / 46 residuals are positive (LEPA EQR higher than our LRMI), confirming a small systematic downward bias on boundary-region sites; 43 / 46 have |residual| 0.08, i.e. essentially boundary near-misses.

Output: `Outputs/9_per_metric_diagnostic.xlsx`.

Note: this script inferred metric responsibility indirectly from percentile rank because LEPA’s per-metric EQRs were not yet available. Script 10 (below) supersedes that indirect inference with a direct LEPA-vs-LT comparison of the same per-metric values.

10.3 LEPA per-metric comparison (script 10)

Script: `10_lepa_per_metric_comparison.R`. LEPA’s published per-metric EQRs (DSFI, ASPT, #DEP, %EHP-%CrHi) plus the final LRMI / class are read from `Inputs/LEPA metrics.xlsx` (82 curated sites) and joined to our LT-OTL run on `site_code`. For each metric the script reports the per-site residual (LEPA – LT-OTL), its mean / median / |mean|, Pearson r, and the percentage of samples agreeing to within ± 0.005 .

Outputs:

- `Outputs/10_lepa_per_metric_comparison.xlsx` — four sheets: per-site residuals (`per_site`), raw summary (`summary`), and two publication-ready tables (`publication_distribution`, `publication_residuals`).
- `Plots/12_lepa_per_metric_panel.tiff` — combined A/B panel: five-panel scatter on top (LEPA vs LT-OTL per metric, with 1:1 reference and per-facet Pearson r / p / n) and five-panel boxplot on the bottom (LEPA vs OTL distributions per metric, jittered points overlaid, WFD class bands on the EQR facet).

This is the definitive comparison: it lets the Discussion attribute residuals directly to a specific metric rather than inferring it.

11 Findings (use to anchor the Results / Discussion text)

These are the headline numerical results the pipeline produces. They are the most important values to verify before publication.

1. **2013 is the outlier year, not OTL standardisation.** A per-year breakdown of |LEPA EQR – LT-OTL LRMI| (over 1026 site-years with a valid LEPA EQR) shows 2013 with mean |residual| 0.085 and median 0.057, against 2014–2022 mean |residual| of 0.011–0.023 and median 0.000–0.010. Interpretation: LEPA’s pre-2014 EQRs were calculated under the old single-DSFI assessment, not the LRMI. Filtering to 2014–2022 produces clean comparison data.
2. **DEP is not the bias source.** Per the diagnostic in §8.1, DEP-implied LRMI shifts correlate only $r = 0.17$ with the actual residual. Most of the residual is not explained by OTL collapsing E+P species.
3. **No single metric dominates the bias on boundary sites (indirect inference).** Per §8.2, even the most-often-low metric (EHP-CrHi) accounts for only 37% of the 46 worst sites. The bias is distributed across all four metrics. Note: this finding was derived before LEPA’s per-metric EQRs were available. The direct head-to-head comparison in §8.3 (script 10) provides the definitive per-metric attribution and should be reported in preference once its numbers are pinned down.
4. **95% of LRMI-era site-years (2014–2022) agree closely with LEPA.** Median residual 0.01, mean |residual| 0.014–0.023. The remaining ~5% are mostly small class-boundary near-misses.
5. **Per-metric residuals against LEPA — EHP-CrHi carries the gap.** Direct comparison against LEPA’s published per-metric EQRs on the 82-site curated subset (script 10) gives:

Metric	n	Mean residual(LEPA – LT-OTL)	Mean residual	Pearson r	% within ± 0.005
DSFI	82	–0.008	0.016	0.981	90.2%
ASPT	82	+0.001	0.005	0.996	68.3%
#DEP	82	+0.005	0.005	0.998	93.9%
%EHP	82	+0.062	0.062	0.774	35.4%
—					
%CrHi					
LRMI (final EQR)	82	+0.015	0.018	0.991	43.9%

Interpretation. Three of the four per-metric EQRs (DSFI, ASPT, #DEP) agree with LEPA at $r = 0.98$ and mean |residual| 0.016 — essentially identical. The residual on the final LRMI is driven almost entirely by %EHP – %CrHi: mean residual +0.062 (LEPA higher), mean |residual| 0.062, and the only metric with r below 0.99 ($r = 0.774$, 35% within ± 0.005). The systematic positive sign means LEPA’s EHP-CrHi values run slightly higher than ours, plausibly because the qualitative hand-picked count of crustaceans/hirudineans differs between the two pipelines (LEPA’s treatment of hand-picked specimens is not documented in the published data dictionary).

12 Decisions worth flagging in Materials & Methods

The following choices deviate from a strict reading of the cited papers and should be stated explicitly:

1. **LRMI strict NA rule** — paper averages remaining metrics if DSFI fails; we set LRMI to NA. Effect: **0** samples in our dataset (DSFI always keys out in our data), but worth declaring.
2. **LRMI clamped at 1 and rounded to 2 dp** — paper allows values > 1 (their Fig. 5). We match LEPA’s reported precision and ceiling.
3. **Uniform class boundaries** across all river types — type-specific offsets proposed in the paper (Table 5 italic rows) are not used.
4. **Natural-waterbodies-only scope** — heavily modified waterbodies are excluded from the recalculation.
5. **Oligochaeta collapse for BMWP** — all oligochaete OTUs are pooled into a single scoring unit at score 1, by OTL_taxonname match (Oligochaeta Gen. sp., Naididae Gen. sp., Tubificidae Gen. sp.).
6. **Ancylus as a genus-level BMWP unit** — preserves the historical Ancylidae score distinct from Planorbidae despite the recent taxonomic synonymy.
7. **fwe.info comparison removed** — earlier versions of the pipeline compared OTL against a freshwater ecology.info standardisation; this has been dropped from the current paper scope.
8. **Year filter 2014–2022** — applied from Stage 4 onwards. LEPA’s pre-2014 EQRs were not the LRMI but the older single-DSFI assessment, so including them in the comparison would conflate methodological era with taxonomic-standardisation effects (see Findings §10.1).
9. **Hand-picked abundance treatment** — Stage 4 coerces hand-picked abundances to 1 before aggregating with kick counts. The hand-picked sub-sample is qualitative (intended to extend the taxon list, not provide density estimates), so contributing it as presence-only prevents the qualitative supplement from biasing the abundance-weighted metrics.
10. **Non-indigenous species (NIMS)** — invasive taxa are not given any special treatment in this pipeline. The raw taxonomy is too unreliable for a confident NIMS flag, and the OTL collapses many of the most error-prone groups to coarser units anyway; any NIMS correction at this stage would be downstream of the very uncertainty it is trying to address. Sidagytė-Copilas & Arbačiauskas (2022) flag this as a future-work item and apply no NIMS correction themselves; we follow that convention.
11. **BMWP scores** — the per-family BMWP sensitivity scores are taken from the `BMWP_look up` sheet in `Inputs/All_sites_macroinvertebrate_data_long.xlsx`, which is the single source of truth for the project. No external published table is consulted at run time.
12. **Year filter as a hard methodological cut** — the **2014–2022** window is treated as the “operational LRMI era” and is a hard methodological choice. No supplementary comparison over the 2013 transition is reported, because the goal is a like-for-like comparison of OTL standardisation against LEPA’s LRMI assessment, not a comparison spanning the change from the pre-2014 single-DSFI assessment.
13. **Sample-level residual reported as a continuous value** — script 7’s per-sample comparison table reports `eqr_diff` as a signed continuous residual and `eqc_match` as a boolean class-agreement flag. The earlier coarse magnitude bands (`identical` / `minor` / `moderate` / `large` at 0.05 and 0.10 cutoffs) have been dropped because the thresholds had no external anchor; the continuous residual plus the EQC-match flag captures the same information without imposing arbitrary categorical boundaries.

13 Reproducibility notes

- R version: 4.0.5 (matching the cited paper’s environment).
- Packages: tidyverse, readxl, writexl, vegan, reshape2, ggplot2, sf, ggrepel, cowplot, rnaturalearth, rgeoboundaries, ggspatial.
- Working directory at run time: the project root containing Inputs/, Outputs/, Scripts/.
- Scripts are numbered 1–10 matching the producing stage; their outputs are written with the same numeric prefix in Outputs/ (or Plots/). Older versioned scripts (_v1, _v17 etc.) have been retired to Backup scripts/ and are not part of the live pipeline.
- The .gitignore excludes Backup scripts/, Literature/, and Claude-related artefacts (.claude/, CLAUDE.md) from version control.

14 Open questions for the author

All items previously flagged here have been resolved during pipeline construction and moved into Decisions §9–13. This section is retained as a placeholder for any new items that surface during manuscript review.